Name:…… Combination:……

P525/ 3
Chemistry
(Practical)
Paper 3
Mar/April, 2022
3 ¼ hours.

S.6 EXAMINATIONS – 2022
CHEMISTRY PRACTICAL
PAPER 3
TIME: 3 ¼ HOURS

INSTRUCTIONS TO CANDIDATES

- *Answer **all** questions*
- *Record your answers in this question paper in the spaces provided.*
- *You are not allowed to use any reference book*
- *Mathematical tables, slide rulers and silent non-programmable electronic calculators may be used.*
- *Candidates are not allowed to start working with the apparatus for the first 15 minutes. This time is to ensure that they have all the chemicals and apparatus they need.*

For Examiner's Use Only			
Q1	***Q2***	***Q3***	***TOTAL***

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Question one

You are provided with the following:

HA1; which is a 0.02M potassium manganate(VII) solution.
HA2; which is a 0.1M sodium hydroxide solution.
0.5M sulphuric acid solution.
Solid **M** which is a mixture of sodiumethanedioate and ethanedioic acid.
You are required;

- To determine the percentage composition of the mixture using sodium hydroxide and potassium manganate(VII) solution respectively.

Procedure
Accurately weigh 2.0g of **M**, dissolve it in about 100cm^3 of distilled water and transfer the mixture into a 250cm^3 volumetric flask. Add distilled water up to the mark and label the resultant solution **HA3**.
Mass of weighing dish and sample **M**=……..g
Mass of empty weighing dish =……..g
Mass of sample **M** alone =……..g
PART A
Pipette 25cm^3 or20cm^3 of **HA3** into a clean conical flask followed by 2-3 drops of phenolphthalein indicator and titrate the mixture with **HA2** from the burette until the end point is reached. Repeat the titration until you obtain consistent results and record your results in the table I below.
Volume of pipette used……cm^3

Table I

Experimental run	1	2	3
Final burette reading(cm^3)			
Initial burette reading(cm^3)			
Volume of **HA2** used(cm^3)			

Titre values used to calculate the average volume of **HA2** used ……, ……..cm^3
Average volume of **HA2**
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(a) Write the equation for the reaction that took place.
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(b) Calculate the;
(i) Number of moles of sodium hydroxide solution that reacted. (01 mark)
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(ii) Number of moles of ethanedioate ions that reacted. (01 mark)
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PART B

Pipette 25cm^3 or20cm^3 of **HA3** into a clean conical flask, add an equal volume of 0.5M sulphuric acid and heat the mixture to about 70°C. Titrate the hot solution immediately with HA1 from the burette until the purple solution turns colourless. Repeat the titration to obtain consistent results and record your results in the table II below.

Volume of pipette used……cm^3

Table II

Run	1	2	3
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of **HA1** used (cm^3)			

(a) (i) Titre values used to calculate the average volume of **HA1** used …… cm^3

(ii) Average volume of **HA1**

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(b) Calculate the;

(i) Number of moles of manganate (VII) ions that reacted. (01 mark)

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(ii) Number of moles of ethanedioate ions reacted with manganate (VII) ions (03 marks)

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(c) Calculate the mass of sodium ethanedioate in the mixture. Hence the percentage of ethanoic acid in the mixture.[$Na = 23, C = 12, O = 16, H = 1$] (03$\frac{1}{2}$marks)

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Question two

You are provided with substance E which consists of two cations and two anions. Carry out the following tests to identify the cations and anions in E. identify any gas(es) evolved and record your results in the table below.

TEST	OBSERVATIONS	DEDUCTIONS
(a) To a little of E, add $8cm^3$ of water. Filter and keep both the filtrate and residue. Divide the filtrate into 5 portions.		
(i) To the first portion, add dilute sodium hydroxide solution dropwise until in excess.		
(ii) To the second portion, add ammonia solution dropwise until in excess.		

(iii) To the third portion, carry out a test of your choice to confirm the cation in the filtrate. …… …… …… ……..		
(iv) To the fourth portion, add lead ethanoate followed by ethanoic acid.		
(v) To the fifth portion, carry out a test of your own choice to identify the anion in the filtrate. …… …… …… ……..		
(b) Wash the residue with water and add dilute nitric acid. Divide the solution into 3 portions.		
(i) To the 1st portion, add dilute sodium hydroxide solution until in excess.		
(ii) To the 2nd portion, add ammonia solution dropwise until in excess.		
(iii) To the 3rd portion, carry out a test of your own choice to confirm the cation in the residue.		

(c) Identify the cations and anions in E.

Cations: ……..

Anions: ……..

Question three

You are provided with organic compound **N** which contains one functional group. You are required to determine the functional group and comment on the nature of N. where a gas(es) evolved must be identified. Record your observations and deductions in the table below.

TESTS	OBSERVATIONS	DEDUCTIONS
(a) Burn a small amount of N on a spatula end.		
(b) To about $1cm^3$ of N in a testtube add about 2 cm^3 of water and shake. Then test the resultant solution with litmus paper.		
(c) To about $0.5cm^3$of N, add about $1cm^3$of acidified potassium dichromate solution and heat.		
(d) To about $0.5cm^3$of N, add 3 cm^3 of Brady's reagent.		
(e) To about $0.5cm^3$ of N, add 2 cm^3 of Iodine solution followed by dropwise addition of sodium hydroxide solution and warm.		
(f) To about $0.5cm^3$ of N in a clean testtube, add Tollen's reagent and warm.		

Comment on the nature of N

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END